Background & Academic Pathway

I have joined the Ph.D. program in Construction Management Technology (Spring 2026) with a fairly non-traditional background. I started in Civil Engineering with a major in Geotechnical Engineering, which gave me a strong technical foundation. However, I realized early on that purely technical skills can become very niche. I wanted the leverage to understand problems at a larger scale, to grasp system components and business logic before diving into solutions.

That drive led me to pursue an MBA in Finance, which helped me learn how to interconnect different disciplines. Later, I added an MS in Computer Information Systems to build hard skills in Big Data and Artificial Intelligence. Now, I am returning to a Ph.D. because I want to synthesize these areas. My goal is to use my background in modeling, robotics, and large language models to solve industry problems where traditional engineering methods often fall short.

Research Interests

I am currently starting a research project on spatiotemporal data in construction. This is not necessarily my long-term research focus. Instead, I see it as a training ground—a way to navigate the Ph.D. process, engage deeply with the literature, and understand foundational research workflows.

A Ph.D. requires a specific rhythm: identifying a problem, thinking about it deeply, and determining whether it is solvable. I am using this initial work to build that competency. Eventually, I plan to pivot toward applying AI/ML and robotics to address specific friction points in the construction industry, such as automated progress monitoring or predictive risk modeling.

Professional Aspirations & Challenges

Ideally, I want to stay connected to academia. While I enjoy the pace of industry, I have always found teaching deeply rewarding. I previously lectured in mathematics and sciences, and I found the experience fulfilling. My ideal scenario would involve working in industry to remain engaged with real-world problems while also serving as a visiting or resident professor.

The biggest challenge for me is not execution; once I know a problem is solvable, I can complete the work. The difficulty lies in the ambiguity beforehand: identifying the right problem and the appropriate methodology. I also maintain varied interests outside of research. Rather than being a distraction, this balance helps me maintain mental health and a fresh perspective.

Individual Development Plan (IDP): Stated Goals

Goal 1: Academic Publications

I aim to submit my current work on spatiotemporal data to a major journal by the end of this summer. In the long term, my goal is to publish at least five journal articles and five conference papers before graduation.

Goal 2: Technical Competency

I want to ensure my technical skills remain sharp and relevant to the computer science job market. This includes mastering deep neural networks, computer vision, and time-series analysis. I also plan to solve at least 300 LeetCode problems and obtain a professional cloud certification.

Goal 3: Industry Experience

Beginning in my second year, I plan to pursue multiple co-op and internship opportunities, remaining continuously engaged with industry through graduation.